

Homework 2
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Question 1:

A = adjacency matrix V_1 = column vector of ones

a)

$$\begin{aligned}k_i &= \sum_{j=1}^N A_{ij} \\AV_1 &= \sum_{j=1}^N A_{ij}(V_1) \\&= \sum_{j=1}^N A_{ij}(1) \\&= \sum_{j=1}^N A_{ij} \\&= k_i \\ \mathbf{k} &= AV_1\end{aligned}$$

b)

$$\begin{aligned}2m &= \sum_{i=1}^N k_i \\m &= \frac{1}{2} \sum_{i=1}^N k_i \\m &= \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N A_{ij}\end{aligned}$$

c)

$$(A^2)_{ij} = \sum_{k=1}^N A_{ik}A_{kj}$$

If k is a common neighbor of i and j , then $A_{ik}A_{kj} = 1$. $\therefore N_{ij} = (A^2)_{ij}$.

d)

$T = \#$ of triangles

$A^3 = \#$ of length-three walks

$$\text{Tr}(A^3) = \sum_{i=1}^N (A^3)_{ii}$$

Each triangle is counted six times $\therefore$ divide by six

$$T = \frac{1}{6} \text{Tr}(A^3)$$

e)

i) k, degrees = [5 5 5 4 5 2 6 2 1 1 2 2 2]

ii) m, edges = 21

iii) common neighbors for 0,3 = 3

iv) triangles = 18